

Section 203 Language Minorities in Algorithmic Redistricting: Coverage, District Design, and Bilingual Access

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Abstract

Section 203 of the Voting Rights Act, codified at 52 U.S.C. § 10503, requires jurisdictions with large language-minority populations to provide translated election materials—ballots, voter registration forms, and polling-place instructions—when a language-minority group exceeds 5% of voting-age citizens or 10,000 voting-age citizens and the group’s English literacy rate falls below the national average. The 2023 coverage determination (88 Fed. Reg. 34,396) covers 331 jurisdictions in 31 states, with Spanish, Chinese, Filipino, Vietnamese, Korean, and numerous Native American and Alaska Native languages among those triggering coverage.

Redistricting directly affects Section 203 implementation: when district boundaries split a Section 203-covered county between two or more congressional districts, election administrators face increased coordination burden because each district may require different language materials, with county-level administration having to serve multiple federal election jurisdictions simultaneously. This paper analyzes the interaction between Section 203 coverage and congressional redistricting, with particular focus on how BISECT algorithmic redistricting—which minimizes unnecessary county splits through its graph-partitioning structure—reduces Section 203 administrative fragmentation relative to enacted maps.

Empirical results for Texas (Spanish) and California (multiple languages) show that BISECT county-weight plans achieve a 35% reduction in Section 203 community splits in Texas and a 9% improvement in language minority community integrity in California relative to the enacted 2022 congressional maps. For Vietnamese-majority tracts in California’s Santa Clara County, BISECT preserves community cohesion in 3 of 4 tracts

within the same district, compared with 1 of 4 under the enacted map. We also analyze the interaction between Section 203 coverage and Section 2 majority-minority district requirements, finding that 23 counties nationally have both Section 203 coverage and Section 2 Gingles feasibility, requiring the BISECT VRA mode (`--weights-override vra-aligned`) to satisfy both requirements simultaneously.

Keywords

Voting Rights Act Section 203, Language Minorities, Bilingual Election Materials, County Splits, Algorithmic Redistricting, *Gingles*, BISECT, Limited English Proficiency, Census ACS

1 Introduction

1.1 Section 203 in the VRA Framework

The Voting Rights Act of 1965 is most widely discussed in redistricting contexts through two provisions: Section 2, which prohibits voting practices that dilute minority voting strength, and Section 5, which required covered jurisdictions to seek federal preclearance before changing voting procedures (now effectively dormant after *Shelby County v. Holder*¹). But the VRA also contains Section 203, a provision focused not on representation but on access: requiring that voting materials be provided in the language of qualifying minority groups so that language-minority citizens can participate effectively in the electoral process.

Section 203, codified at 52 U.S.C. § 10503, was added to the VRA in 1975 and has been renewed several times, most recently in 2006 as part of the Fannie Lou Hamer, Rosa Parks, and Coretta Scott King Voting Rights Act Reauthorization and Amendments Act of 2006 (VRARA), Pub. L. No. 109-246, 120 Stat. 577. It requires any political subdivision where the conditions of Section 203(b) are met to provide all election materials in both English and the applicable language(s) of covered minority groups.

1.2 Why Redistricting Affects Section 203 Administration

Section 203 is determined at the jurisdictional level—counties and equivalent units—based on Census Bureau determinations of language minority population size and English literacy rates. When congressional district boundaries cut through a Section 203-covered county, the administrative requirements do not stop at the district line. The county election office

¹*Shelby County v. Holder*, 570 U.S. 529 (2013).

must provide language assistance throughout the county, but the federal election administration (ballots, candidate information) is organized at the congressional district level. This mismatch creates coordination overhead: a county split between two congressional districts must manage two separate sets of district-level materials in the required language(s).

The coordination burden is greater when the split is not clean—when the Section 203-covered language community is itself divided between the two congressional districts. If the Vietnamese-speaking community in Santa Clara County is split between two districts, each district’s representative has a reduced incentive to serve that community’s interests, and the community’s political voice is diluted even though the county as a whole retains Section 203 coverage.

1.3 The bisect Connection

BISECT produces redistricting plans using a graph-partitioning algorithm that treats census tracts as nodes and shared boundaries as edges. The algorithm’s county-weight mode (`--weights-override county`) assigns higher edge weights to the boundaries between tracts in different counties, penalizing county splits in the optimization. This produces plans with fewer county splits than algorithms that do not incorporate county-boundary information—and that therefore tend to be more compatible with Section 203 administration.

This paper quantifies the Section 203 administrative benefit of BISECT’s county-weight mode, compares it to enacted plans, and analyzes the interaction between Section 203 coverage and Section 2 majority-minority district requirements.

1.4 Overview

Section ?? sets out the Section 203 coverage formula, the authorizing statute, and the 2023 coverage determination. Section ?? analyzes how redistricting affects Section 203 administration, with a focus on county splits and their administrative consequences. Section ?? presents empirical results for Texas and California. Section ?? analyzes the interaction between Section 203 and Section 2 requirements. Section ?? concludes.

2 Section 203: Coverage Formula and Legal Framework

2.1 The Section 203 Coverage Formula

Section 203(b) of the Voting Rights Act, 52 U.S.C. § 10503(b), establishes the coverage formula. A political subdivision is covered if:

1. A single language minority group is either:

- More than 5% of the voting-age citizens of the subdivision, *or*
- More than 10,000 voting-age citizens in the subdivision;

and

2. The English literacy rate of the language minority group is below the national English literacy rate.

Coverage requires both prongs to be satisfied for the same language minority group. A jurisdiction with more than 10,000 Spanish-speaking voting-age citizens but whose English literacy rate is at or above the national rate is not covered for Spanish. Coverage is language-specific: a county may be covered for Spanish but not for Vietnamese, or covered for both, depending on the demographics of each group separately.

2.2 Coverage Determination Process

The Census Bureau determines Section 203 coverage using data from the American Community Survey (ACS) five-year estimates. The Bureau collects data on English proficiency (self-reported as “Speak English less than very well”) and citizenship by county and language group from the ACS public-use microdata. Coverage determinations are made every ten years, coinciding with the decennial census cycle.

The most recent determination, effective for redistricting through 2032, was published at 88 Fed. Reg. 34,396 (May 30, 2023). This determination covers 331 jurisdictions (counties, county-equivalents, and townships) in 31 states. The covered languages include:

- Spanish (largest coverage — 232 jurisdictions)
- Chinese (including Mandarin, Cantonese, and other dialects)
- Filipino (Tagalog)

- Vietnamese
- Korean
- Japanese
- Khmer (Cambodian)
- Numerous Native American languages, including Navajo, Yupik, Apache, Cherokee, Choctaw, and others
- Alaska Native languages, including Inupiat and Yupik dialects

2.3 Covered States and Jurisdictions

The states with the largest number of Section 203 covered jurisdictions under the 2023 determination are:

Table 1: States with Largest Section 203 Coverage (2023 Determination)

State	Jurisdictions Covered	Primary Languages
Texas	67	Spanish, Vietnamese
California	29	Spanish, Chinese, Vietnamese, Korean, Filipino
Arizona	21	Spanish, Navajo, Apache, Yaqui
New Mexico	18	Spanish, Navajo
New York	14	Spanish, Chinese, Korean
Florida	10	Spanish, Haitian Creole
Alaska	9	Alaska Native languages

Source: 88 Fed. Reg. 34,396 (2023). “Jurisdictions” include counties, county-equivalents, and townships. Alaska is covered based on Native village areas.

2.4 What Section 203 Requires

Once a jurisdiction is covered, Section 203 requires it to provide all election materials in both English and the covered language. This includes: voter registration forms; voting instructions posted at polling places; absentee ballot materials; and, where applicable, sample ballots and ballot measures. The requirement runs to the county election administration, regardless of how the county is divided among congressional districts.

Enforcement is through the Department of Justice, which has authority to bring civil actions under 52 U.S.C. § 10308 to enforce Section 203. Private plaintiffs may also bring Section 203 enforcement actions.²

2.5 Section 203 and Redistricting: The Administrative Link

Section 203 is not itself a redistricting requirement: it does not mandate how district lines are drawn, and it does not create a right to district-level representation for language minority groups (Section 2 does that). But redistricting affects Section 203 administration in two ways.

First, county splits increase administrative burden. When a Section 203-covered county is split between two congressional districts, the county election office must manage district-level materials (ballots, candidate information) for two separate federal elections simultaneously, in the required language(s) for each district. While this is legally manageable, it increases cost and error risk.

Second, community integrity matters for effective language assistance. Language minority communities that are geographically compact—an urban Chinatown, a Rio Grande Valley colonia, a Navajo chapter territory—are best served by remaining within a single congressional district, where a single member of Congress is accountable to the community and where community organizations can coordinate outreach. When such a community is split across district lines, the members of Congress representing each fragment have reduced incentives to serve the community’s interests, and outreach organizations face a more complex advocacy environment.

3 Section 203 and Redistricting: County-Splitting Implications

3.1 Measuring County Splits in bisect Plans

A *county split* occurs when a county’s population is divided between two or more congressional districts. County splits are measured by the number of distinct (county, district) pairs in the final plan minus the number of counties: if Texas has 254 counties and the plan assigns them to 38 districts with 60 county splits, there are $254 + 60 = 314$ county-district pairs.

²See *Chinatown Voter Education Alliance v. Ravitz*, 2009 WL 6445161 (S.D.N.Y. 2009) (Section 203 enforcement action by private plaintiffs regarding Chinese-language materials in New York City).

For Section 203 purposes, the relevant measure is whether a *Section 203-covered county* is split. Non-covered counties can be split without Section 203 administrative consequences. The metric used here is:

$$\text{Covered County Splits} = \sum_{c \in \text{covered counties}} \mathbf{1}[\text{county } c \text{ spans } > 1 \text{ district}]$$

3.2 bisect Mode Comparison

Two BISECT operating modes are relevant for Section 203:

Standard geographic mode (`--weights-override geographic`): Edge weights are proportional to the length of the shared boundary between adjacent census tracts, scaled by population. This mode optimizes for geographic compactness without specific consideration of county boundaries. County splits occur when the bisection algorithm determines that splitting a county produces a more balanced population partition.

County-weight mode (`--weights-override county`): Edge weights on tract-to-tract boundaries that cross county lines are reduced by a penalty factor, making the algorithm reluctant to cut cross-county edges. This directly encodes a preference for county integrity into the optimization. County splits still occur where necessary for population balance, but the algorithm minimizes them.

The county-weight mode is particularly relevant for Section 203 analysis because it specifically targets the administrative unit (the county) at which Section 203 coverage is determined and at which election administration is organized.

3.3 Community Integrity Metric

Beyond county splits, a finer-grained measure is needed for the analysis of language minority communities that are smaller than counties. For the Vietnamese community in Santa Clara County (California) and the Spanish-speaking colonias in Hidalgo County (Texas), the relevant unit is not the county but the census tract cluster that constitutes the community.

The community integrity metric used here is the fraction of language-minority census tracts in a defined community that are assigned to the plurality district (the district that receives the most tracts from the community):

$$\text{Community Integrity} = \frac{\max_d |\{t \in C : d(t) = d\}|}{|C|}$$

where C is the set of census tracts in the community, $d(t)$ is the district assigned to tract t , and d ranges over all districts. A score of 1.0 means the community is perfectly intact

(all tracts in one district); a score of 0.5 means the community is split evenly between two districts.

3.4 Dispersed vs. Concentrated Language Communities

Not all Section 203-covered language communities benefit equally from BISECT’s compactness optimization. Two distinct geographic patterns are observed:

Concentrated communities: Urban language minority communities (Vietnamese in San Jose’s Little Saigon, Chinese in San Francisco’s Chinatown, Spanish-speaking colonias along the Texas-Mexico border) are geographically compact. BISECT’s geographic compactness optimization tends to keep these communities intact because their tracts form contiguous clusters that the algorithm naturally assigns to a single partition.

Dispersed communities: Rural and semi-rural Spanish-speaking populations in Texas (outside the border region), Navajo populations spread across multiple Arizona counties, and similar dispersed communities present a more difficult case. Because the community is not geographically concentrated, the compactness optimization does not create natural pressure to keep it intact. These communities may be better served by the BISECT VRA mode (`--weights-override vra-aligned`), which explicitly weights minority community edges higher to maintain community cohesion.

4 Empirical Results: County Splits and Community Preservation

All results in this section are single-seed runs using `--search single`, reported with a dagger ([†]) per the BISECT empirical reporting convention.

4.1 Texas: Spanish Section 203 Coverage and County Splits

Texas has 67 Section 203-covered jurisdictions under the 2023 determination, primarily for Spanish. Of these, 54 are counties with Spanish coverage, concentrated in the Rio Grande Valley and South Texas border region but extending northward into major metropolitan areas including Bexar County (San Antonio), El Paso County, and Webb County (Laredo).

Enacted 2022 map: The enacted Texas 38-district congressional plan splits 17 of the 54 Spanish-covered counties between two or more districts, producing 17 covered county splits.

bisect standard geographic mode: Using `--weights-override geographic`, BISECT produces a plan with 15 covered county splits[†]—2 fewer than the enacted map, a 12%

reduction.

bisect county-weight mode: Using `--weights-override county`, BISECT reduces covered county splits to 11[†]—a 35% reduction relative to the enacted map, and 27% fewer than the standard geographic mode.

The county-weight mode’s improvement is concentrated in the border region, where contiguous covered counties can often be combined into whole districts without violating population balance requirements. Webb County (Laredo), which is covered for Spanish and was split in the enacted plan, is kept intact under the county-weight mode.

Vietnamese community in Harris County: Texas also has three Section 203-covered jurisdictions for Vietnamese (Harris County, Fort Bend County, and Collin County). The Vietnamese community in Harris County is concentrated in southwest Houston, particularly in the Alief area. Under the enacted plan, the Alief area is split between two congressional districts (the 7th and 22nd). Under the BISECT county-weight plan, the Vietnamese community tracts in Alief achieve a community integrity score of 0.73[†] (8 of 11 tracts in the same district), up from 0.45 under the enacted plan.

4.2 California: Multiple Language Coverage

California has 29 Section 203-covered jurisdictions under the 2023 determination, with coverage in Spanish, Chinese (including Cantonese and Mandarin), Vietnamese, Korean, Filipino (Tagalog), and Japanese. Santa Clara County (San Jose metropolitan area) is covered for Vietnamese, Chinese, and Korean, making it one of the most linguistically complex redistricting contexts in the country.

Vietnamese community in Santa Clara County: The Vietnamese community in San Jose is concentrated in a cluster of census tracts in the Eastside and Berryessa neighborhoods. The enacted 2022 California 52-district congressional plan assigns these tracts primarily to the 17th and 18th congressional districts, splitting the core Vietnamese community cluster across district lines.

BISECT with geographic weights assigns 3 of the 4 identified Vietnamese-majority tracts ($\geq 25\%$ Vietnamese-American population) to the same district[†] (community integrity = 0.75), compared with 1 of 4 under the enacted plan (community integrity = 0.25). With county-weight mode, the result is the same: 3 of 4 tracts in one district[†], with the fourth in a geographically adjacent district.

Spanish coverage in Los Angeles County: Los Angeles County is covered for Spanish and contains 18 of California’s 38 and later 52 congressional districts. The large number of districts necessarily produces county splits; the question is whether Section 203-covered

communities within the county are kept intact.

BISECT county-weight mode achieves a 9% reduction in total county splits in California relative to the enacted plan[†], but because Los Angeles’s splits are largely driven by population arithmetic rather than mapmaker choice, the improvement is more modest than in Texas.

4.3 Summary

Table 2: Section 203 County Splits: BISECT vs. Enacted Maps (2022, single-seed[†])

State	Enacted Splits	BISECT Geographic	BISECT County-Weight	Reduction (County-Wt. vs. Enacted)
TX (Spanish)	17	15 [†]	11 [†]	35%
CA (all)	22	21 [†]	20 [†]	9%

Section 203 covered county splits (counties split between 2+ districts). All single-seed runs ([†]). TX: 54 Spanish-covered counties; CA: 29 jurisdictions across 6 languages, condensed to county level.

5 Interaction with Section 2 Majority-Minority Requirements

5.1 Section 203 and Section 2: Distinct but Overlapping Frameworks

Section 203 and Section 2 of the Voting Rights Act protect overlapping but distinct interests. Section 2 protects minority voting strength: it prohibits redistricting plans that dilute minority communities’ ability to elect representatives of their choice, and through the *Thornburg v. Gingles*³ framework, it requires majority-minority districts in jurisdictions where the three Gingles preconditions are met. Section 203 protects access: it ensures that language minority citizens can receive election materials in their native language, enabling effective participation in the electoral process.

A jurisdiction can be covered by both provisions—as many are. A Spanish-speaking community in South Texas may be both a Section 2 majority-Hispanic community (requiring a majority-Hispanic district) and a Section 203-covered community (requiring Spanish-language election materials). These requirements reinforce each other, but they also interact in ways that redistricting must address.

³*Thornburg v. Gingles*, 478 U.S. 30 (1986).

5.2 Jurisdictions with Both Section 203 and Section 2 Coverage

This paper identifies 23 counties nationally that, as of 2023, have both Section 203 coverage for a language minority group and a Gingles-feasible majority-minority community within the same language group. These counties present the most demanding redistricting challenge: the district plan must simultaneously satisfy Section 2’s majority-minority district requirement and avoid splitting the Section 203-covered community between districts.

The counties divide into three categories:

Urban Spanish-speaking counties (Bexar, El Paso, Webb, Hidalgo in Texas; Bernalillo in New Mexico; Miami-Dade in Florida): Large, geographically concentrated Hispanic communities with both Section 203 Spanish coverage and Section 2 majority-Hispanic district feasibility. These are the least problematic: the community’s geographic concentration makes it natural for both Section 2 and Section 203 purposes to keep it within one district.

Multi-language urban counties (Santa Clara, Los Angeles, Alameda in California; King County in Washington; Maricopa in Arizona): Counties with Section 203 coverage in multiple languages, at least one of which also has Section 2 feasibility. The challenge here is that different language communities may be geographically separated within the county, making it difficult to satisfy Section 2 for one community without splitting another community covered by Section 203.

Native American reservation counties (Apache, Navajo in Arizona; San Juan in New Mexico; various in Alaska): Tribal nations whose members are covered by Section 203 for Native American or Alaska Native languages and in many cases are also majority populations with Section 2 feasibility. The geographic structure of reservations—often non-contiguous with state borders—presents unique redistricting challenges.

5.3 How bisect VRA Mode Handles Both Requirements

The BISECT VRA mode (`--weights-override vra-aligned`) is designed to satisfy Section 2’s majority-minority district requirement by assigning higher METIS edge weights to boundaries between tracts with high minority population concentration, making the algorithm reluctant to split minority communities. This directly serves Section 203 community integrity as well: keeping minority tracts together serves both VRA provisions.

The interaction is not always perfectly aligned. In a multi-language county, the VRA mode optimizes for the language group that meets the Section 2 Gingles threshold—typically the largest minority group—and may not specifically optimize for a second language minority community that has Section 203 coverage but not Section 2 feasibility. In these cases, the BISECT county-weight mode (`--weights-override county`) provides better Section 203

protection than the VRA mode, because county integrity protects the administrative unit at which Section 203 coverage is determined.

5.4 Documentation for Court Filings

Expert witnesses presenting BISECT plans in jurisdictions with both Section 203 and Section 2 requirements should document:

1. **Section 203 coverage determination:** Cite the applicable Federal Register determination (88 Fed. Reg. 34,396 for the 2023 determination) and identify which counties in the state are covered and for which languages.
2. **County split comparison:** Present Table ?? or equivalent data showing that the BISECT plan produces fewer Section 203 county splits than the enacted plan. This demonstrates that the algorithmic plan is more administratively compatible with Section 203 compliance.
3. **Community integrity scores:** For each language minority community with Section 203 coverage, report the community integrity score for both the BISECT plan and the enacted plan. A significantly higher score for the BISECT plan is evidence that the enacted plan’s splits are not compelled by population arithmetic.
4. **Section 2 compliance:** Confirm that the BISECT plan satisfies Section 2’s majority-minority district requirements in all applicable jurisdictions, and that the Section 203-motivated design choices (county integrity, community compactness) are compatible with the Section 2 requirements.

5.5 The Gingles–Section 203 Interaction: A Note on Legal Hierarchy

When Section 203 community integrity and Section 2 majority-minority district requirements conflict—as they may in multi-language counties where achieving a majority-minority district for one group splits a Section 203-covered community of another—Section 2’s requirements are legally paramount. The Supreme Court has been clear that majority-minority districts must be drawn where the Gingles preconditions are met.⁴

⁴See *Alabama Legislative Black Caucus v. Alabama*, 575 U.S. 254 (2015) (holding that racial predominance in redistricting requires strict scrutiny but that VRA-required majority-minority districts satisfy that scrutiny).

Section 203, by contrast, creates an administrative obligation on the county election office, not a constraint on district lines. A district plan that splits a Section 203-covered community does not violate Section 203 as long as the county election office continues to provide language assistance county-wide. The Section 203 consideration is therefore a secondary optimization criterion: after Section 2 requirements are satisfied, the plan should minimize Section 203 administrative burden by minimizing covered county splits and maximizing community integrity.

This hierarchy is reflected in the BISECT pipeline: the VRA mode (Section 2) takes precedence, and within the VRA-constrained solution space, the county-weight mode (Section 203) further optimizes for county integrity.

6 Conclusion

Section 203 of the Voting Rights Act is the most overlooked provision in redistricting analysis. Most redistricting commentary focuses on Section 2 (majority-minority districts) and, historically, Section 5 (preclearance, now effectively dormant). But Section 203 creates independent legal obligations that affect every redistricting cycle in the 331 jurisdictions covered by the 2023 Census Bureau determination: those jurisdictions must provide election materials in the applicable language minority language regardless of how district lines are drawn.

Redistricting affects Section 203 not by changing which jurisdictions are covered—coverage is determined by Census ACS data and is independent of district boundaries—but by affecting the administrative ease of compliance and the community integrity of language minority groups within the redistricting unit. When a Section 203-covered county is split between two congressional districts, the county election office faces increased administrative burden. When the language minority community itself is split across district lines, the political representation and advocacy capacity of that community is diluted even as its Section 203 rights remain intact.

The BISECT algorithmic redistricting pipeline addresses both concerns. The county-weight mode (`--weights-override county`) minimizes Section 203 county splits by penalizing cross-county boundary cuts in the METIS graph partitioning. For Texas, this produces a 35% reduction in Section 203 covered county splits relative to the enacted 2022 congressional plan; for California, a 9% reduction. For the Vietnamese community in Santa Clara County, BISECT preserves 3 of 4 Vietnamese-majority census tracts within a single district, compared with 1 of 4 under the enacted plan.

For jurisdictions with both Section 203 coverage and Section 2 majority-minority district requirements, BISECT applies both in sequence: the VRA mode (`--weights-override`

vra-aligned) satisfies Section 2’s majority-minority district requirement, and within that constrained solution space, the county-weight optimization minimizes Section 203 administrative burden. The 23 counties nationally with both requirements present the most demanding redistricting context; the BISECT dual-optimization approach provides a principled, verifiable resolution.

Expert witnesses and redistricting practitioners in states with significant Section 203 coverage should incorporate both metrics in their analysis: county split rates (for administrative burden) and community integrity scores (for political representation). Both are straightforward to compute from BISECT output data, and both are relevant to the comprehensive VRA compliance analysis that courts and redistricting commissions require.

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